Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\sin^{-1}(x/5) + \text{cosec}^{-1}(5/4) = \pi/2$ then the value of x is

(A) 1

(B) 3

(C) 4

(D) 5

Q.2

If $\sin^{-1}(x-1) + \cos^{-1}(x-3) + \tan^{-1}\left(\frac{x}{2-x^2}\right)$
$= \cos^{-1} k + \pi$, then the value of $k =$

(A) 1

(B) $-1/\sqrt{2}$

(C) $1/\sqrt{2}$

(D) None of these

Q.3

Simplified form of $\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}\frac{a}{b}\right)$ is

(A) 0

(B) $\frac{2a}{b}$

(C) $\frac{2b}{a}$

(D) $\frac{\pi}{2}$

Q.4

What is the value of :
$\cos\left[\tan^{-1}\left\{\tan\left(\frac{15\pi}{4}\right)\right\}\right]$?

(A) $-\frac{1}{\sqrt{2}}$

(B) 0

(C) $\frac{1}{\sqrt{2}}$

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(D) $\frac{1}{2\sqrt{2}}$

Q.5

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$, then

(A) $x^2 + y^2 + z^2 + xyz = 0$

(B) $x^2 + y^2 + z^2 + 2xyz = 0$

(C) $x^2 + y^2 + z^2 + xyz = 1$

(D) $x^2 + y^2 + z^2 + 2xyz = 1$

Q.6

If $\cos^{-1} \lambda + \cos^{-1} \mu + \cos^{-1} \gamma = 3\pi$, then the value of $\lambda\mu + \mu\gamma + \gamma\lambda$ is

(A) 0

(B) 1

(C) 3

(D) 6

Q.7

If n be the number of distinct solutions of the equation $\cos^{-1}x + \cos^{-1}2x = \pi$, then the value of $\frac{1}{n}$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Q.8

The equation $\tan^{-1}(1+x) + \tan^{-1}(1-x) = \frac{\pi}{2}$ is satisfied by

(A) $x = 1$

(B) $x = -1$

(C) $x = 0$

(D) $x = \frac{1}{2}$

Q.9

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The range of the function
$f(x) = \sin^{-1}(\log[x]) + \log\left(\sin^{-1}[x]\right)$; (where [.] denotes the greatest integer function) is

(A) R

(B) $[1, 2)$

(C) $\left\{\log \frac{\pi}{2}\right\}$

(D) $\{-\sin 1\}$

Q.10

If $\sin^{-1} \frac{1}{x} = \sin^{-1} \frac{1}{a} + \sin^{-1} \frac{1}{b}$, then the value of x is

(A) $\frac{ab}{\sqrt{a^2-1}+\sqrt{b^2-1}}$

(B) $\frac{ab}{\sqrt{a^2-1}-\sqrt{b^2-1}}$

(C) $\frac{2ab}{\sqrt{a^2-1}+\sqrt{b^2-1}}$

(D) None of these

Q.11

The set of values of x for which the identity $\cos^{-1} x + \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right) = \frac{\pi}{3}$ holds good is

(A) $[0, 1]$

(B) $\left[0, \frac{1}{2}\right]$

(C) $\left[\frac{1}{2}, 1\right]$

(D) $\{-1, 0, 1\}$

Q.12

What is the value of $\tan\left(\tan^{-1} x + \tan^{-1} y + \tan^{-1} z\right) - \cot\left(\cot^{-1} x + \cot^{-1} y + \cot^{-1} z\right)$?

(A) 0

(B) $2(x + y + z)$

(C) $\frac{3\pi}{2}$

(D) $\frac{3\pi}{2} + x + y + z$

Q.13

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If $\sin^{-1} a + \sin^{-1} b + \sin^{-1} c = \pi$, then find the value of $a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2}$.

(A) abc

(B) $a + b + c$

(C) $\frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}$

(D) $2abc$

Q.14

The value of $3\tan^{-1}\frac{1}{2} + 2\tan^{-1}\frac{1}{5} + \sin^{-1}\frac{142}{65\sqrt{5}}$ is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) π

(D) None of these

Q.15

If $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$ then the value of q is equal to

(A) 1

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

Q.16

If $\sin^{-1} x + \sin^{-1} y = \pi/2$ and $\cos^{-1} x - \cos^{-1} y = 0$ then values x and y are respectively

(A) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$

(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, -\frac{1}{2}$

(D) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Q.17

$\theta = \tan^{-1}\left(2\tan^2\theta\right) - \tan^{-1}\left(\frac{1}{3}\tan\theta\right)$ then $\tan\theta =$

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(A) -2

(B) -1

(C) $2/3$

(D) 2

Q.18

Total number of positive integral value $'n'$ so that the equations $\cos^{-1} x + \left(\sin^{-1} y\right)^2 = \frac{n\pi^2}{4}$ and $\left(\sin^{-1} y\right)^2 - \cos^{-1} x = \frac{\pi^2}{16}$ are consistent, is equal to

(A) 1

(B) 4

(C) 3

(D) 2

Q.19

Two angles of a triangle are $\cot^{-1} 2$ and $\cot^{-1} 3$. Then the third angle is

(A) $\frac{\pi}{4}$

(B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{3}$

Q.20

If $\cot\left(\cos^{-1} x\right) = \sec\left(\tan^{-1} \frac{a}{\sqrt{b^2-a^2}}\right)$, then x is equal to

(A) $\frac{b}{\sqrt{2b^2-a^2}}$

(B) $\frac{a}{\sqrt{2b^2-a^2}}$

(C) $\frac{\sqrt{2b^2-a^2}}{a}$

(D) $\frac{\sqrt{2b^2-a^2}}{b}$

Answers & Solutions

Q.1 Answer:

3

Solution:

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$\sin^{-1}(x/5)+\operatorname{cosec}^{-1}(5/4)=\pi/2$
$\Rightarrow \sin^{-1}(x/5)+\sin^{-1}(4/5)=\pi/2$
$\Rightarrow \quad \sin^{-1}(x/5)+\cos^{-1}\sqrt{1-16/25}=\pi/2$
$\Rightarrow \sin^{-1}(x/5)+\cos^{-1}(3/5)=\pi/2$
$\Rightarrow x=3$ **as** $\sin^{-1}x+\cos^{-1}x=\pi/2 \quad \forall x.$

Q.2 Answer:

$1/\sqrt{2}$

Solution:

$\sin^{-1}(x-1) \Rightarrow -1 \le x-1 \le 1 \Rightarrow 0 \le x \le 2$
$\cos^{-1}(x-3) \Rightarrow -1 \le x-3 \le 1 \Rightarrow 2 \le x \le 4$
$\tan^{-1}\left(\frac{x}{2-x^2}\right) \Rightarrow x \in R, x \neq \sqrt{2}, -\sqrt{2}$
$\sin^{-1}(2-1)+\cos^{-1}(2-3)+\tan^{-1}\frac{2}{2-4}=\cos^{-1}k+\pi$
$\Rightarrow \sin^{-1}1+\cos^{-1}(-1)+\tan^{-1}(-1)=\cos^{-1}k+\pi$
$\frac{\pi}{2}+\pi-\frac{\pi}{4}=\cos^{-1}k+\pi$
$\Rightarrow \cos^{-1}k=\frac{\pi}{4} \Rightarrow k=\frac{1}{\sqrt{2}}$

Q.3 Answer:

$\frac{2b}{a}$

Solution:

Let $\frac{1}{2}\cos^{-1}\frac{a}{b}=\theta$**; then**
$\cos^{-1}\frac{a}{b}=2\theta; \Rightarrow \cos 2\theta=\frac{a}{b}$ **then expression**
$=\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)$
$=\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}$
$=\frac{(1+\tan\theta)^2+(1-\tan\theta)^2}{(1-\tan\theta)(1+\tan\theta)}$
$=\frac{2+2\tan^2\theta}{1-\tan^2\theta}=\frac{2\left(1+\tan^2\theta\right)}{1-\tan^2\theta}$
$=\frac{2\left(\cos^2\theta+\sin^2\theta\right)}{(\cos^2\theta-\sin^2\theta)}$
$=\frac{2}{\cos 2\theta}=\frac{2}{\frac{a}{b}}=\frac{2b}{a}$

Q.4 Answer:

$\frac{1}{\sqrt{2}}$

Solution:

The given trigonometric expression is :
$\cos\left[\tan^{-1}\left\{\tan\left(\frac{15\pi}{4}\right)\right\}\right]$
$=\cos\left[\tan^{-1}\left\{\tan\left(4\pi-\frac{\pi}{4}\right)\right\}\right]$
$=\cos\left[\tan^{-1}\left\{-\tan\frac{\pi}{4}\right\}\right]=\cos\left[\tan^{-1}\tan\left(\frac{-\pi}{4}\right)\right.$
Since, $\tan^{-1}\theta$ **is defined for** $\frac{-\pi}{2}<\theta<\frac{-\pi}{2}$

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$= \cos\left(\frac{-\pi}{4}\right)$

$\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ $[\text{since } \cos(-\theta) = \cos\theta]$

Q.5 Answer:

$x^2 + y^2 + z^2 + 2xyz = 1$

Solution:

Given that $\cos^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(z) = \pi$

$\Rightarrow \cos^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(z) = \cos^{-1}(-1)$

$\Rightarrow \cos^{-1}(x) + \cos^{-1}(y) = \pi - \cos^{-1}(z)$

$\Rightarrow \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) = \cos^{-1}(-z)$

$\Rightarrow xy - \sqrt{(1-x^2)(1-y^2)} = -z$

$\Rightarrow (xy + z) = \sqrt{(1-x^2)(1-y^2)}$

Squaring both sides, we get $x^2 + y^2 + z^2 + 2xyz = 1.$

Q.6 Answer:

3

Solution:

We know that $0 \le \cos^{-1} x \le \pi$. **Hence, from the question** $\cos^{-1}\lambda = \pi, \cos^{-1}\mu = \pi, \cos^{-1}\gamma = \pi$

$[\because \cos^{-1}\lambda + \cos^{-1}\mu + \cos^{-1}\gamma = 3\pi$ **is possible only when each term attains its maximum.]** $\Rightarrow \lambda = \mu = \gamma = -1 \Rightarrow \lambda\mu + \mu\gamma + \gamma\lambda = 3$

Q.7 Answer:

1

Solution:

$x \geq 0$

$\cos^{-1}x + \cos^{-1}2x = \pi.$

$2x = \cos\pi - \cos^{-1}x$

$= -\cos\cos^{-1}x$

$= -x \quad \Rightarrow \quad x = 0$

$x \leq 0$

$\cos^{-1}-x + \cos^{-1}-2x = \pi$

$\pi - \cos^{-1}x + \pi - \cos^{-1}2x = \pi$

$\cos^{-1}x + \cos^{-1}2x = \pi$

Q.8 Answer:

$x = 0$

Solution:

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$\tan^{-1}(1+x)+\tan^{-1}(1-x)=\frac{\pi}{2}$
$\tan^{-1}\left[\frac{(1+x)+(1-x)}{1-(1+x)(1-x)}\right]=\frac{\pi}{2}$
$\Rightarrow \frac{1+x+1-x}{1-(1+x)(1-x)}=\tan\frac{\pi}{2}$
$\Rightarrow \frac{2}{1-(1+x)(1-x)}=\frac{1}{0}\Rightarrow 1-(1+x)(1-x)=0$
$\Rightarrow (1+x)(1-x)=1$
$1-x^2=1$
$x^2=0$
$x=0$

Q.9 Answer:

$\left\{\log\frac{\pi}{2}\right\}$

Solution:

$\sin^{-1}(\log[x])$ **is defined if** $-1\le\log[x]\le 1$
and $[x]>0$
$\Rightarrow \frac{1}{e}\le[x]\le e\Rightarrow[x]=1,2\Rightarrow x\in[1,3)$
Again, $\log\left(\sin^{-1}[x]\right)$ **is defined if**
$\sin^{-1}[x]>0$ **and** $-1\le[x]\le 1$
$\Rightarrow [x]>0$ **and** $-1\le[x]\le 1\Rightarrow 0<[x]\le 1$
$\Rightarrow x\in[1,2)$
$\therefore$ **Domain of** $f(x)=[1,2)$ **For** $1\le x<2,[x]=1$
$\therefore \quad f(x)=\sin^{-1}0+\log\frac{\pi}{2}=\log\frac{\pi}{2},\forall x\in[1,2)$
$\therefore$ **Range of** $f(x)=\left\{\log\frac{\pi}{2}\right\}$

Q.10 Answer:

$\frac{ab}{\sqrt{a^2-1}+\sqrt{b^2-1}}$

Solution:

Let $\sin^{-1}\frac{1}{a}=\theta;\sin^{-1}\frac{1}{b}=\phi$ **then**
$\sin^{-1}\frac{1}{x}=\theta+\phi$
$\Rightarrow \sin\sin^{-1}\frac{1}{x}=\sin(\theta+\phi)$
$\Rightarrow \frac{1}{x}=\sin\theta\cos\phi+\cos\theta\sin\phi$
$=\frac{1}{a}\sqrt{1-\frac{1}{b^2}}+\sqrt{1-\frac{1}{a^2}}\cdot\frac{1}{b}=\frac{\sqrt{b^2-1}}{ab}+\frac{\sqrt{a^2-1}}{ab}$
$\Rightarrow x=\frac{ab}{\sqrt{a^2-1}+\sqrt{b^2-1}}$

Q.11 Answer:

$\left[\frac{1}{2},1\right]$

Solution:

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Case 1:

If $0 \leq x \leq \frac{1}{2}$, **then** $\cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)$

$\cos^{-1}\left(x \times \frac{1}{2}+\sqrt{1-x^2}\frac{\sqrt{3}}{2}\right)$

$=\cos^{-1} x-\cos^{-1}\frac{1}{2}$

Therefore, the equation is

$\cos^{-1} x+\cos^{-1} x-\cos^{-1}\frac{1}{2}=\frac{\pi}{3} \Rightarrow x=\frac{1}{2}$

Case 2 : If $\frac{1}{2} \leq x \leq 1$, **then**

$\cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)=\cos^{-1}\frac{1}{2}-\cos^{-1} x$

Therefore, the equation is

$\cos^{-1} x+\cos^{-1}\frac{1}{2}-\cos^{-1} x=\frac{\pi}{3}$

which is an identity. Hence, the identity holds good for $x \in \left[\frac{1}{2}, 1\right]$.

Q.12 Answer:

0

Solution:

$\tan\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)-$
$\cot\left(\cot^{-1} x+\cot^{-1} y+\cot^{-1} z\right)$
$=\tan\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)-$
$\cot\left(\frac{\pi}{2}-\tan^{-1} x+\frac{\pi}{2}-\tan^{-1} y+\frac{\pi}{2}-\tan^{-1} z\right)$
$\left(\because \tan^{-1}\mathrm{x}+\cot^{-1}\mathrm{x}=\frac{\pi}{2}\right)$
$=\tan\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)$
$-\cot\left\{3\pi/2-\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)\right.$
$=\tan\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)$
$-\tan\left(\tan^{-1} x+\tan^{-1} y+\tan^{-1} z\right)=0$

Q.13 Answer:

$2abc$

Solution:

Let $\sin^{-1} a=x \quad \therefore a=\sin x$
$\sin^{-1} b=y \quad \therefore b=\sin y; \sin^{-1} c=z$
$\therefore c=\sin z$
$\therefore \quad a\sqrt{1-a^2}+b\sqrt{1-b^2}+c\sqrt{1-c^2}$
$=\sin x \cos x+\sin y \cos y+\sin z \cos z$
$=(1/2)(\sin 2x+\sin 2y+\sin 2z)=(1/2)(4 \sin x \sin y \sin z)$
$=2 \sin x \sin \mathrm{y} \sin z==2\mathrm{abc}$

Q.14 Answer:

π

Solution:

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We have,

$3\tan^{-1}\frac{1}{2}+2\tan^{-1}\frac{1}{5}+\sin^{-1}\frac{142}{65\sqrt{5}}$

$=2\left(\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}\right)+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{142}{31}$

$=2\tan^{-1}\frac{7}{9}+\pi+\tan^{-1}\frac{\frac{1}{2}+\frac{142}{31}}{1-\left(\frac{1}{2}\right)\left(\frac{142}{31}\right)}$

$=\tan^{-1}\frac{\frac{14}{9}}{1-\frac{49}{81}}+\pi-\tan^{-1}\frac{315}{80}$

$=\pi-\tan^{-1}\frac{63}{16}+\tan^{-1}\frac{63}{16}=\pi$

Q.15 Answer:

$\frac{1}{2}$

Solution:

Let $\cos^{-1}\sqrt{p}+\cos^{-1}\sqrt{1-p}$

$+\cos^{-1}\sqrt{1-q}=\frac{3\pi}{4}$

Let $a=\cos^{-1}\sqrt{p} b=\cos^{-1}\sqrt{1-p}$ and

$c=\cos^{-1}\sqrt{1-q}$

$\Rightarrow \cos a=\sqrt{p}, \cos b=\sqrt{1-p}, \cos c=\sqrt{1-q}$

$\Rightarrow \cos^2 a=p, \cos^2 b=1-p, \cos^2 c=1-q$

Now, $\sin^2 a=1-\cos^2 a=1-p$

$\Rightarrow \sin a=\sqrt{1-p}$,

$\sin^2 b=1-\cos^2 b=1-1+p \Rightarrow \sin b=\sqrt{p}$

$\sin^2 c=1-\cos^2 c=1-1+q=q \Rightarrow \sin c=\sqrt{q}$

$\therefore$ equation (i) can be written as $a+b+c=\frac{3\pi}{4} \Rightarrow a+b=\frac{3\pi}{4}-c$

Take cos on each side, we get $\cos(a+b)=\cos\left(\frac{3\pi}{4}-c\right)$

$\Rightarrow \cos a\cos b-\sin a\sin b$

$=\cos\left\{\pi-\left(\frac{\pi}{4}+c\right)\right\}=-\cos\left(\frac{\pi}{4}+c\right)$

Put values of $\cos a, \cos b$ and $\sin a, \sin b$, we get

$\sqrt{p}\cdot\sqrt{1-p}-\sqrt{1-p}\sqrt{p}$

$=-\left(\frac{1}{\sqrt{2}}\sqrt{1-q}-\frac{1}{\sqrt{2}}\sqrt{q}\right)$

$\Rightarrow \quad 0=\sqrt{1-q}-\sqrt{q} \Rightarrow \sqrt{1-q}=\sqrt{q}$

Squaring on both side: $\Rightarrow 1-q=q \Rightarrow \quad 1=2q \Rightarrow q=\frac{1}{2}$

Q.16 Answer:

$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Solution:

Given, $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$

and $\cos^{-1}x-\cos^{-1}y=0$

$\Rightarrow \left(\frac{\pi}{2}-\sin^{-1}x\right)-\left(\frac{\pi}{2}-\sin^{-1}y\right)=0$

$\Rightarrow \sin^{-1}y-\sin^{-1}x=0 \Rightarrow \sin^{-1}y=\sin^{-1}x$

From equations (i) and (ii), we get

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$2\sin^{-1}x = \frac{\pi}{2}$

$\Rightarrow \sin^{-1}x = \frac{\pi}{4} \Rightarrow x = \frac{1}{\sqrt{2}}$

From equation (ii)

$y = \frac{1}{\sqrt{2}}$

Q.17 Answer:

-2

Solution:

$\theta = \tan^{-1}\left[\frac{2\tan^2\theta - \frac{1}{3}\tan\theta}{1+\frac{2}{3}\tan^3\theta}\right]$

$\Rightarrow \tan\theta = \frac{6\tan^2\theta - \tan\theta}{3+2\tan^3\theta} \Rightarrow 1 = \frac{6\tan\theta - 1}{3+2\tan^3\theta}$

$\Rightarrow 2\tan^3\theta - 6\tan\theta + 4 = 0$

$\Rightarrow (\tan\theta - 1)^2(\tan\theta + 2) = 0$

$\Rightarrow \tan\theta = 1; \tan\theta = -2; \tan\theta = 0.$

Q.18 Answer:

1

Solution:

$$\cos^{-1}x + \left(\sin^{-1}y\right)^2 = \frac{n\pi^2}{4}$$

$$-\cos^{-1}x + \left(\sin^{-1}y\right)^2 = \frac{\pi^2}{16}$$

$$2\left(\sin^{-1}(y)^2 = \frac{\pi^2}{4}\left(n + \frac{1}{4}\right)\right.$$

$$\frac{-\pi}{2} \le \sin^{-1}y \le \frac{\pi}{2}$$

$$0 \le \left(\sin^{-1}y\right)^2 \le \frac{\pi^2}{4}$$

$$0 \le 2\sin^{-1}y\Big)^2 \le \frac{\pi^2}{2}$$

$$0 \le \frac{\pi^2}{4}\left(n + \frac{1}{4}\right) \le \frac{\pi^2}{2}$$

$$0 \le n + \frac{1}{4} \le 2$$

$$-\frac{1}{4} \le n \le \frac{7}{4}$$

$n = 1$ **only the positive Integer**

Q.19 Answer:

$\frac{3\pi}{4}$

Solution:

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The sum of the two given angles is $\cot^{-1}(2) + \cot^{-1}(3) = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$
$= \tan^{-1}(1) = \frac{\pi}{4}$
So, the third angle is $\pi - \frac{\pi}{4} = \frac{3\pi}{4}$

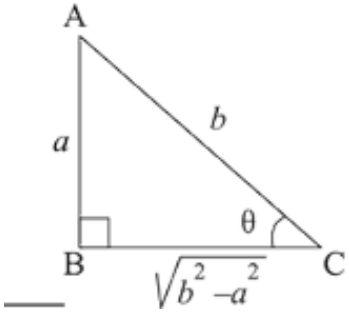
Q.20 Answer:

$\frac{b}{\sqrt{2b^2-a^2}}$

Solution:

$\cot\left(\cos^{-1} x\right) = \sec\left(\tan^{-1}\frac{a}{\sqrt{b^2-a^2}}\right) \cdots$ **(i)**

Let $\theta = \tan^{-1}\frac{a}{\sqrt{b^2-a^2}}$



$\therefore \tan\theta = \frac{a}{\sqrt{b^2-a^2}}$

In $\triangle ABC, AC^2 = a^2 + \left(\sqrt{b^2-a^2}\right)^2$

$AC = b$

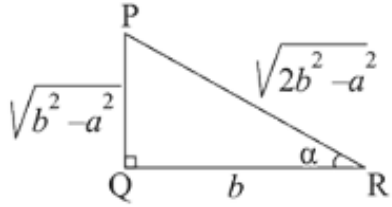
$\therefore \sec\theta = \frac{b}{\sqrt{b^2-a^2}}$

From (i), $\cot\left(\cos^{-1} x\right) = \sec\theta$

$\cot\left(\cos^{-1} x\right) = \frac{b}{\sqrt{b^2-a^2}}$ **[using (ii)]**

$\cos^{-1} x = \cot^{-1}\left(\frac{b}{\sqrt{b^2-a^2}}\right)$ **....(iii)**

Again let $\alpha = \cot^{-1}\left(\frac{b}{\sqrt{b^2-a^2}}\right)$



$\cot\alpha = \frac{b}{\sqrt{b^2-a^2}}$

$\therefore PR^2 = \left(\sqrt{b^2-a^2}\right)^2 + b^2$

$PR = \sqrt{2b^2-a^2}$

$\therefore \cos\alpha = \frac{b}{\sqrt{2b^2-a^2}}$ **... (iv)**

From(iii), $\cos^{-1} x = \alpha$ **or** $x = \cos\alpha$

$x = \frac{b}{\sqrt{2b^2-a^2}}$ **[using (iv)]**

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